Lecture 26 Cluster Analysis III

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

1. Introduction
2. Summary and Remark

Maximum likelihood estimation (MLE)

Example (maximum likelihood estimation, motivation)

Suppose that you had just two observations $x_1 = 6.5$, $x_2 = 7.5$ from a normal population whose variance is 1 and the mean is known to be either 7, or 10. What would you choose as an estimate of the population mean, and why?

We define the "likelihood", a quantity that measures the degree to which something is likely to happen, by the joint density function (or the joint probability mass function, in case of discrete random variables) values at the observations.

$$L(\mu) = p(x_1|\mu) \times p(x_2|\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_1-\mu)^2}{2}\right) \times \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_2-\mu)^2}{2}\right)$$

Therefore, $L(7)$ measures that if the real mean $\mu = 7$, to what degree would the observations $x_1 = 6.5$ and $x_2 = 7.5$ be likely to happen; and $L(10)$ gives the degree to which the same observation is likely to happen. A simple computation shows that $L(7) = 0.12395$ and $L(10) = 1.52966 \times 10^{-5}$. it makes more sense to choose $\hat{\mu} = 7$.

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Definition (likelihood function)

Let $X_1, X_2, ..., X_n$ be a random sample from a distribution characterized by $p(x|\theta)$, where θ is an unknown parameter. Let $x_1, \ldots, x_n$ be the observed values. The likelihood function is defined by

$$L(\theta) = L(\theta|x_1, x_2, ..., x_n) = \prod_{i=1}^{n} p(x_i|\theta).$$

For the sake of simplicity in applications, people also use

Definition (log-likelihood function)

$$l(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p(x_i|\theta).$$

We see that if one parameter θ maximizes $L(\theta)$, it also maximizes $l(\theta)$.
The density function $p(x|\theta)$ is determined by the parameter θ, and should not be taken as the conditional density function with condition θ. The parameter θ

The maximum likelihood estimator is defined below.

Definition (maximum likelihood estimator, **MLE**)

The maximum likelihood estimator $\hat{\theta}_{\mathsf{mle}} = \hat{\theta}_{\mathsf{mle}}(X_1, \ldots, X_n)$ is defined to be the parameter that maximizes the likelihood function $L(\theta)$ (or equivalently, maximizes the log-likelihood function $l(\theta)$).

$$\hat{\theta}_{\mathsf{mle}} = \arg\max_{\theta} l(\theta|\boldsymbol{x}).$$

Remark

Note that the MLE, $\hat{\theta}$ is a function of the sample $\hat{\theta}(X_1, \ldots, X_n)$, and therefore it is a random variable. According to our previous Definition, the MLE $\hat{\theta}$ is a statistic, and is also referred to as a point estimator.

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Remark

If the unknown distribution p is specified by more than one parameters, $p = p(x|\theta_1, \ldots, \theta_k)$, then the likelihood function is defined by

$$L(\theta_1, \ldots, \theta_k) = \prod_{i=1}^{n} p(x_i|\theta_1, \ldots, \theta_k),$$

and the log-likelihood function is defined to be $l(\theta_1, \ldots, \theta_k) = \log L(\theta_1, \ldots, \theta_k)$. The MLE is the vector $(\hat{\theta}_1, \ldots, \hat{\theta}_k)$ that maximizes L or l.

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How to find the MLE

The usual way to finding the maximizer of a function $L(\theta)$ is the following two steps:

1. Find θ such that
$$\frac{d}{d\theta}L(\theta) = 0; \tag{1.1}$$
2. Verify this value of θ indeed is a maximizer by check that for this value of θ,
$$\frac{d^2}{d\theta^2}L(\theta) \leq 0. \tag{1.2}$$

Remark

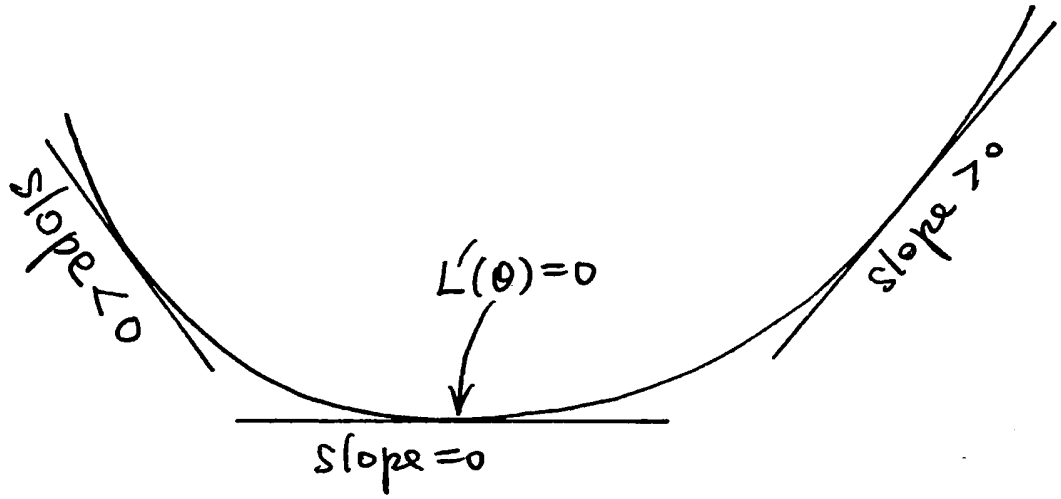
The solution to Equation (1.1) may not be unique.

Remark

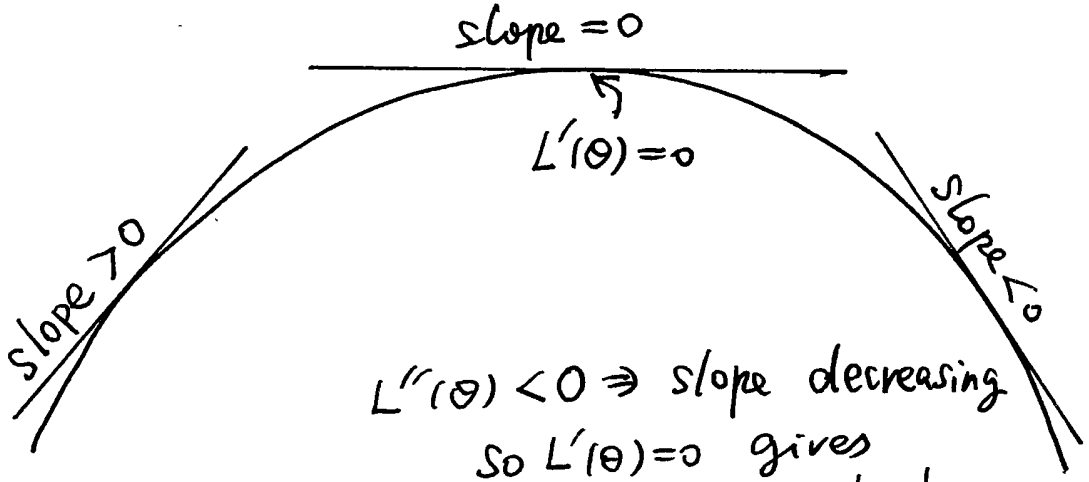
Strictly speaking the above two steps will only find *local* maxima, while the

Remark

In both of the equations (1.1) and (1.2), the function $L(\theta)$ can be changed to $l(\theta) = \log L(\theta)$ since the logarithm function is strictly increasing.



$L''(\theta) > 0 \Rightarrow$ slope increasing so $L'(\theta)=0$ gives minimizer.
NOT MLE



$L''(\theta) < 0 \Rightarrow$ slope decreasing so $L'(\theta)=0$ gives maximizer, which IS MLE

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Example

Let $X_1, \ldots, X_n$ be independently sampled from $N(\theta, 1)$. Find the MLE of θ.

Solution. The log-likelihood function (think, why we do not use likelihood function directly?) is

$$l(\theta) = \log \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_i - \theta)^2}{2}\right) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{n}(\theta - x_i)^2.$$

One takes derivative to give $\frac{dl(\theta)}{d\theta} = -\sum_{i=1}^{n}(\theta - x_i)$. Let $\frac{dl(\theta)}{d\theta} = 0$ to give $\theta = \frac{1}{n}\sum_{i=1}^{n} x_i$. We verify that this is really the maximizer by looking at the second order derivative $\frac{d^2 l(\theta)}{d\theta^2} = -n < 0$. We replace the observation by the random sample to give the maximum likelihood estimator (MLE),

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}.$$

Numerical Calculation

If you are lucky, $l(\theta|\boldsymbol{x})$ will be differentiable and you will be able to solve

$$\frac{\partial l(\theta|\boldsymbol{x})}{\partial \theta} = 0$$

in closed form for a θ that maximizes $l(\theta|\boldsymbol{x})$. However, in most cases, maximizing $l(\theta|\boldsymbol{x})$ is a complicated problem that does not have a closed form solution.

- **Coordinate ascent**

$$\theta_1^{(k+1)} = \arg\max_{\theta_1} l(\theta_1, \theta_2^{(k)}|\boldsymbol{x}),$$
$$\theta_2^{(k+1)} = \arg\max_{\theta_2} l(\theta_1^{(k+1)}, \theta_2|\boldsymbol{x}).$$

- **Newton-Raphson**

$$\theta^{(k+1)} = \theta^{(k)} - \frac{l'}{l''}.$$

- **Expectation-maximization, the EM algorithm**

Conditional Probability

Let u_1 and u_2 be two random vectors with joint probability density $p(u_1, u_2)$ and marginal density $p(u_2) = \int p(u_1, u_2) du_1$ for u_2. Then the conditional density function $f(u_1|u_2)$ given i_2, is given by

$$p(u_1|u_2) = \frac{p(u_1, u_2)}{p(u_2)}.$$

Example

Let $u_1, u_2 \in \mathbb{R}$. Consider the density function

$$p(u_1, u_2) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{u_1^2 + u_2^2}{2\sigma^2}\right\}.$$

In this example the marginal density of u_2 is given by

$$p(u_2) = \int_{-\infty}^{\infty} p(u_1, u_2) du_1 = \frac{1}{2\pi\sigma^2} e^{-\frac{u_2^2}{2\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{u_1^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{u_2^2}{2\sigma^2}}.$$

The conditional density $p(u_1|u_2)$ given u_2 is given by $p(u_1|u_2) = \frac{p(u_1, u_2)}{}$

The EM Algorithm

Let $\boldsymbol{x} = \{x_i\}_{i=1}^m$ be the observed data, and let $\boldsymbol{z} = \{z_i\}_{i=1}^n$ be the latent data (the missing data, or the hidden variable). One has

$$p(\boldsymbol{x}, \boldsymbol{z}|\theta) = p(\boldsymbol{z}|\boldsymbol{x}, \theta)p(\boldsymbol{x}|\theta),$$

or equivalently,

$$p(\boldsymbol{x}|\theta) = \frac{p(\boldsymbol{x}, \boldsymbol{z}|\theta)}{p(\boldsymbol{z}|\boldsymbol{x}, \theta)}.$$

One takes logarithms to give

$$\log p(\boldsymbol{x}|\theta) = \log p(\boldsymbol{x}, \boldsymbol{z}|\theta) - \log p(\boldsymbol{z}|\boldsymbol{x}, \theta).$$

Recall the likelihood and the log-likelihood functions

$$L(\theta|\boldsymbol{x}) = p(\boldsymbol{x}|\theta), \qquad l(\theta|\boldsymbol{x}) = \log p(\boldsymbol{x}|\theta).$$

For any given θ^{old}, each iteration of the EM algorithm finds some θ^{new}, so that

$$l(\theta^{\text{new}}|\boldsymbol{x}) \geq l(\theta^{\text{old}}|\boldsymbol{x}).$$

- In some scenarios EM behaves similar as the gradient ascent methods;
- In some scenarios with hidden variables, direct application of the gradient descent methods can be very difficult.

The EM Algorithm

Recall that for any θ,

$$l(\theta|\boldsymbol{x}) = \log p(\boldsymbol{x}|\theta) = \log p(\boldsymbol{x}, \boldsymbol{z}|\theta) - \log p(\boldsymbol{z}|\boldsymbol{x}, \theta).$$

Suppose $\boldsymbol{z} \sim p(\boldsymbol{z}|x, \theta^{\text{old}})$. We take the conditional expectation of the above equation given $\boldsymbol{x}$, and with respect to θ^{old} to give

$$\begin{aligned} l(\theta|\boldsymbol{x}) &= \int [\log p(\boldsymbol{x}, \boldsymbol{z}|\theta)]\, p(\boldsymbol{z}|\boldsymbol{x}, \theta^{\text{old}}) d\boldsymbol{z} - \int [\log p(\boldsymbol{z}|\boldsymbol{x}, \theta)]\, p(\boldsymbol{z}|\boldsymbol{x}, \theta^{\text{old}}) d\boldsymbol{z} \\ &=: Q(\theta, \theta^{\text{old}}) - H(\theta, \theta^{\text{old}}). \end{aligned}$$

Definition (EM)

$$\theta^{\text{new}} = \arg\max_{\theta} Q(\theta, \theta^{\text{old}}).$$

Jensen's Inequality

Lemma (Jensen's inequality)

If $u''(w) < 0$ *almost surely, then* $\mathrm{E}[u(X)] \leq u(\mathbb{E}[X])$.

Since $u''(w) < 0$, the function is strictly concave. So the graph of u lies below the tangent line to u at any point on the graph. Take the point $(\mu, u(\mu))$ with $\mu = \mathbb{E}[X]$. Then

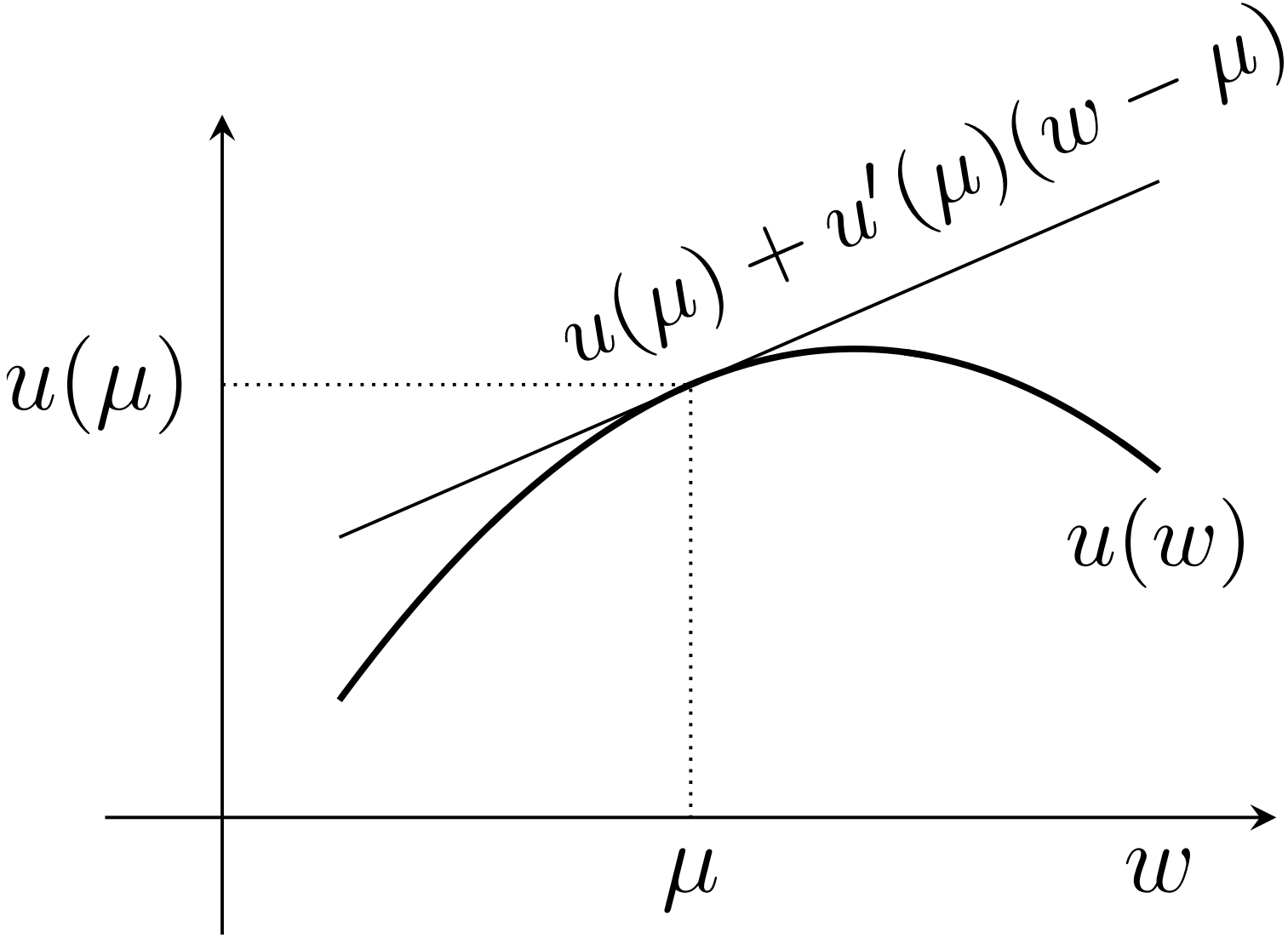
$$u(w) \leq u(\mu) + u'(\mu)(w - \mu), \quad \forall w.$$

Taking expected values on both sides, we have

$$\mathbb{E}[u(X)] \leq u(\mu) + 0 = u(\mathbb{E}[X]).$$

Remark

The equality $\mathbb{E}[u(X)] = u(\mathbb{E}[X])$ holds if and only if X is constant.


$u(\mu) + u'(\mu)(w - \mu)$
$u(\mu)$
$u(w)$
μ
w

We now prove that $l(\theta^{\text{new}}|\boldsymbol{x}) \geq l(\theta^{\text{old}}|\boldsymbol{x})$. In fact, $Q(\theta^{\text{new}}, \theta^{\text{old}}) \geq Q(\theta^{\text{old}}, \theta^{\text{old}})$. Also, by Jensen's inequality,

$$
\begin{aligned}
& H(\theta^{\text{new}}, \theta^{\text{old}}) - H(\theta^{\text{old}}, \theta^{\text{old}}) \\
= & \int \left[\log \frac{p(z|\boldsymbol{x}, \theta^{\text{new}})}{p(z|\boldsymbol{x}, \theta^{\text{old}})} \right] p(z|\boldsymbol{x}, \theta^{\text{old}}) dz \\
\leq & \log \left\{ \int \frac{p(z|\boldsymbol{x}, \theta^{\text{new}})}{p(z|\boldsymbol{x}, \theta^{\text{old}})} p(z|\boldsymbol{x}, \theta^{\text{old}}) dz \right\} \\
= & \log \left\{ \int p(z|\boldsymbol{x}, \theta^{\text{new}}) dz \right\} \\
= & \log 1 = 0.
\end{aligned}
$$

This proves that

$$
\begin{aligned}
& l(\theta^{\text{new}}|\boldsymbol{x}) = Q(\theta^{\text{new}}, \theta^{\text{old}}) - H(\theta^{\text{new}}, \theta^{\text{old}}) \\
& \geq Q(\theta^{\text{old}}, \theta^{\text{old}}) - H(\theta^{\text{old}}, \theta^{\text{old}}) = l(\theta^{\text{old}}|\boldsymbol{x}).
\end{aligned}
$$

Some comments

- From

$$l(\theta|\boldsymbol{x}) = Q(\theta, \theta^{\text{old}}) - H(\theta, \theta^{\text{old}}),$$

  one maximizes $Q(\theta, \theta^{\text{old}})$ (w.r.t. θ) to increase the value of $l(\theta|\boldsymbol{x})$. This is the insight behind the EM algorithm.
- The problem $\max_\theta l(\theta|\boldsymbol{x})$ is decomposed into a sequence of maximization problems

$$\theta^{(k+1)} = \arg\max_\theta Q(\theta, \theta^{(k)}).$$

  In many problems this decomposition greatly simplifies the computation.

Monte Carlo Calculation of $Q(\theta, \theta^{\text{old}})$

In the case that the integral

$$Q(\theta, \theta^{\text{old}}) = \int [\log p(\boldsymbol{x}, \boldsymbol{z}|\theta)]\, p(\boldsymbol{z}|\boldsymbol{x}, \theta^{\text{old}}) dz$$

is difficult to find, the Monte Carlo method is recommended:

- Draw $z_1, \ldots, z_m$ iid from the distribution $p(z|\boldsymbol{x}, \theta^{\text{old}})$;
- Approximate $Q(\theta, \theta^{\text{old}})$ as

$$Q(\theta, \theta^{\text{old}}) \approx \frac{1}{m} \sum_i \log p(z_i, \boldsymbol{x}|\theta).$$

Generalized EM

When the maximization problem

$$\theta^{(k+1)} = \arg\max_{\theta} Q(\theta, \theta^{(k)})$$

is difficult, one may find some $\theta^{(k+1)}$, which does not necessarily maximizes Q, but simply makes

$$Q(\theta^{(k+1)}, \theta^{(k)}) > Q(\theta^{(k)}, \theta^{(k)}).$$

since this is all that is required to enerate a sequence of $\theta^{(k)}$ that increases the log-likelihood, this procedure is useful as well. This procedure is referred to as "generalized EM algorithm" (or simply GEM).

An example.

Suppose that 197 animals (Y) are distributed into four categories as follows

$$Y = (y_1, y_2, y_3, y_4) = (125, 18, 20, 34),$$

with cell probabilities

$$\left(\frac{1}{2} + \frac{\theta}{4}, \frac{1}{4}(1-\theta), \frac{1}{4}(1-\theta), \frac{\theta}{4}\right).$$

We augment the observed data by splittin the first cell into two cells with probabilities $1/2$ and $\theta/4$. The augmented data are given by

$$(x_1, x_2, x_3, x_4, x_5),$$

such that

$$x_1 + x_2 = y_1, \quad x_3 = y_2, \quad x_4 = y_3, \quad x_5 = y_4.$$

We will infer θ from the data.

Method 1: MLE. One has that for any given θ, the vector Y has a multinomial distribution with

$$p(y_1, y_2, y_3, y_4|\theta)$$
$$= \frac{(y_1+y_2+y_3+y_4)!}{y_1!y_2!y_3!y_4!}\left(\frac{1}{2}+\frac{\theta}{4}\right)^{y_1}\left(\frac{1}{4}(1-\theta)\right)^{y_2+y_3}\left(\frac{\theta}{4}\right)^{y_4}.$$

The log-likelihood function is (where C is a constant independent of θ)

$$l(\theta|Y) = \log p(y_1, y_2, y_3, y_4|\theta)$$
$$= C + y_1 \log(2+\theta) + (y_2+y_3)\log(1-\theta) + y_4 \log\theta.$$

Therefore

$$\frac{\partial l(\theta|Y)}{\partial\theta} = \frac{y_1}{2+\theta} - \frac{y_2+y_3}{1-\theta} + \frac{y_4}{\theta}.$$

We see that $\frac{\partial l(\theta|Y)}{\partial\theta}$ is decreasing, and

$$\lim_{\theta\to 0^+}\frac{\partial l(\theta|Y)}{\partial\theta} = \infty, \qquad \lim_{\theta\to 1^-}\frac{\partial l(\theta|Y)}{\partial\theta} = -\infty.$$

So l is minimized by the solution to the equation

$$0 = \frac{\partial l(\theta|Y)}{\partial \theta} = \frac{y_1}{2+\theta} - \frac{y_2+y_3}{1-\theta} + \frac{y_4}{\theta}.$$

The above equation is reduced to

$$(y_1+y_2+y_3+y_4)\theta^2 + (-y_1+2y_2+2y_3+y_4)\theta - 2y_4 = 0.$$

Since $\theta \in (0,1)$, we drop the negative root and obtain the MLE:

$$\hat{\theta}_{\mathsf{mle}} = \frac{1}{2}r + \sqrt{\frac{1}{4}r^2 + \frac{2y_4}{y_1+y_2+y_3+y_4}},$$
$$r = \frac{y_1-2y_2-2y_3-y_4}{y_1+y_2+y_3+y_4}.$$

Method 2: EM. We need to find the funciton Q.

$$Q(\theta, \theta^{\text{old}}) = \int \log p(X, Y|\theta) p(X|Y, \theta^{\text{old}}) dX.$$

First, since Y is only a function of X, and X has multinomial distribution, one has

$$p(X, Y|\theta) = \frac{(\sum x_i)!}{x_1! \cdots x_5!} \left(\frac{1}{2}\right)^{x_1} \left(\frac{\theta}{4}\right)^{x_2} \left(\frac{1-\theta}{4}\right)^{x_3+x_4} \left(\frac{\theta}{4}\right)^{x_5}.$$

Given Y, x_3, x_4, x_5 are fixed, $x_1 = y_1 - x_2$, and $x_1 \sim \text{Binomial}(y_1, \frac{2}{2+\theta})$. So

$$p(X|Y, \theta) = \binom{y_1}{x_1} \left(\frac{2}{2+\theta}\right)^{x_1} \left(\frac{\theta}{2+\theta}\right)^{x_2}.$$

Another way of finding $p(X|Y, \theta)$ is to divide $p(X, Y|\theta)$ by $p(Y|\theta)$, where

$$p(Y|\theta) = \sum_{x_1+x_2=y_1} p(X, Y|\theta)$$

$$
\begin{aligned}
p(Y|\theta) &= \sum_{x_1+x_2=y_1} p(X,Y|\theta) \\
&= \sum_{x_1+x_2=y_1} \frac{(\sum x_i)!}{x_1!\cdots x_5!}\left(\frac{1}{2}\right)^{x_1}\left(\frac{\theta}{4}\right)^{x_2+x_5}\left(\frac{1-\theta}{4}\right)^{x_3+x_4} \\
&= \frac{(\sum y_i)!}{y_1!y_2!y_3!y_4!}\left(\frac{1-\theta}{4}\right)^{y_2+y_3}\left(\frac{\theta}{4}\right)^{y_4}\left(\frac{1}{2}+\frac{\theta}{4}\right)^{y_1} \\
&\quad \times \sum_{x_1+x_2=y_1} \frac{y_1!}{x_1!x_2!}\left(\frac{1/2}{\frac{1}{2}+\frac{\theta}{4}}\right)^{x_1}\left(\frac{\theta/4}{\frac{1}{2}+\frac{\theta}{4}}\right)^{x_2} \\
&= \frac{(\sum y_i)!}{y_1!y_2!y_3!y_4!}\left(\frac{1}{2}+\frac{\theta}{4}\right)^{y_1}\left(\frac{1-\theta}{4}\right)^{y_2+y_3}\left(\frac{\theta}{4}\right)^{y_4}.
\end{aligned}
$$

It is easy to verify that

$$
p(X|Y,\theta) = \frac{p(X,Y|\theta)}{p(Y\theta)}.
$$

Since $x_2|Y \sim \text{Binomial}(y_1, \frac{\theta}{2+\theta})$, one has

$$\mathbb{E}[x_2|Y,\theta] = \frac{\theta y_1}{2+\theta}.$$

$$Q(\theta,\theta^{\text{old}}) = \int \log p(X,Y|\theta)p(X|Y,\theta^{\text{old}})dX = \mathbb{E}[\log p(X,Y|\theta)|Y,\theta^{\text{old}}]$$
$$= \mathbb{E}[C + (x_3+x_4)\log(1-\theta) + (x_2+x_5)\log\theta|Y,\theta^{\text{old}}]$$
$$= \mathbb{E}[C|Y,\theta^{\text{old}}] + (y_2+y_3)\log(1-\theta) + \left(\frac{\theta^{\text{old}}y_1}{2+\theta^{\text{old}}} + y_4\right)\log\theta.$$

Take derivatives to give

$$\frac{\partial Q(\theta,\theta^{\text{old}})}{\partial\theta} = -\frac{y_2+y_3}{1-\theta} + \left(\frac{\theta^{\text{old}}y_1}{2+\theta^{\text{old}}} + y_4\right)\frac{1}{\theta}.$$

Let $\frac{\partial Q(\theta,\theta^{\text{old}})}{\partial\theta} = 0$, the EM iteration takes the form $\theta^{\text{new}} = M(\theta^{\text{old}})$, where

$$M(\theta) = \frac{2y_4 + (y_1+y_4)\theta}{2(y_2+y_3+y_4) + (\sum y_j)\theta}.$$

A brief analysis of convergence: Recall that the true solution θ^* satisfies

$$(y_1 + y_2 + y_3 + y_4)\theta^{*2} + (-y_1 + 2y_2 + 2y_3 + y_4)\theta^* - 2y_4 = 0,$$

or equivalently

$$\theta^* = \frac{2y_4 + (y_1 + y_4)\theta^*}{2(y_2 + y_3 + y_4) + (\sum y_j)\theta^*} = M(\theta^*).$$

One has that for $\theta \in (0, 1)$,

$$\frac{d}{d\theta}\frac{A + B\theta}{C + D\theta} = \frac{B(C + D\theta) - D(A + B\theta)}{(C + D\theta)^2} = \frac{BC - DA}{(C + D\theta)^2},$$

$$|M'(\theta)| \leq \left| \frac{2(y_1 + y_4)(y_2 + y_3 + y_4) - 2y_4(y_1 + y_2 + y_3 + y_4)}{4(y_2 + y_3 + y_4)^2} \right|$$

$$\leq \frac{y_1(y_2 + y_3)}{2(y_2 + y_3 + y_4)^2} = \frac{125 \times 38}{2 \times 72^2} = 0.4581.$$

So

$$|\theta^{\text{new}} - \theta^*| = |M(\theta^{\text{old}}) - M(\theta^*)| = |M'(\theta')(\theta^{\text{old}} - \theta^*)| \leq 0.4581|\theta^{\text{old}} - \theta^*|.$$

Gaussian Mixture Distribution

The Gaussian mixture distribution is defined to be the distribution with the following density

$$p(\boldsymbol{x}) = p(\boldsymbol{x}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k).$$

Here

- $\boldsymbol{x} \in \mathbb{R}^n$, so the Gaussian mixture distribution is supported on $\mathbb{R}^n$.
- $\pi_1, \ldots, \pi_K \in (0, 1)$ and $\pi_1 + \cdots + \pi_K = 1$.
- $\boldsymbol{\mu}_k \in \mathbb{R}^n$ is the mean, and $\boldsymbol{\Sigma}_k \in \mathbb{R}^{n \times n}$ is the covariance matrix, of the k'th Gaussian density function $\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$.
- Parameters: $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)^T$, $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K)$, and $\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_K)$.

Example of Gaussian mixture distribution in 1-dim: we set $K = 2$, $\boldsymbol{\mu} = (-2, 2)$, and $\boldsymbol{\Sigma} = (1, 1)$.

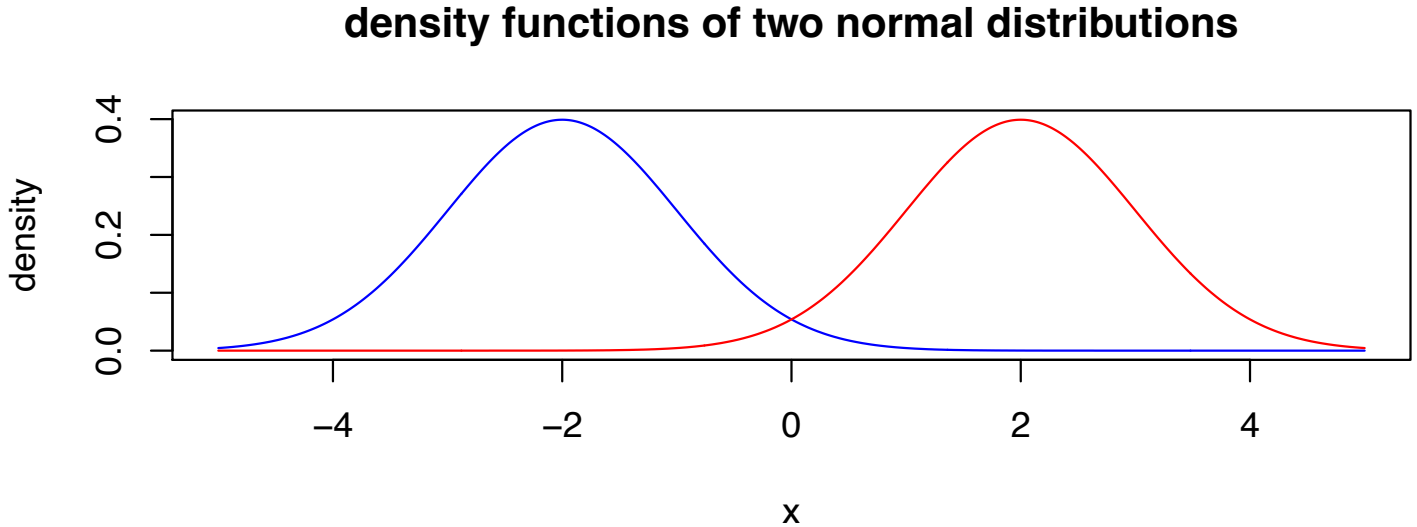
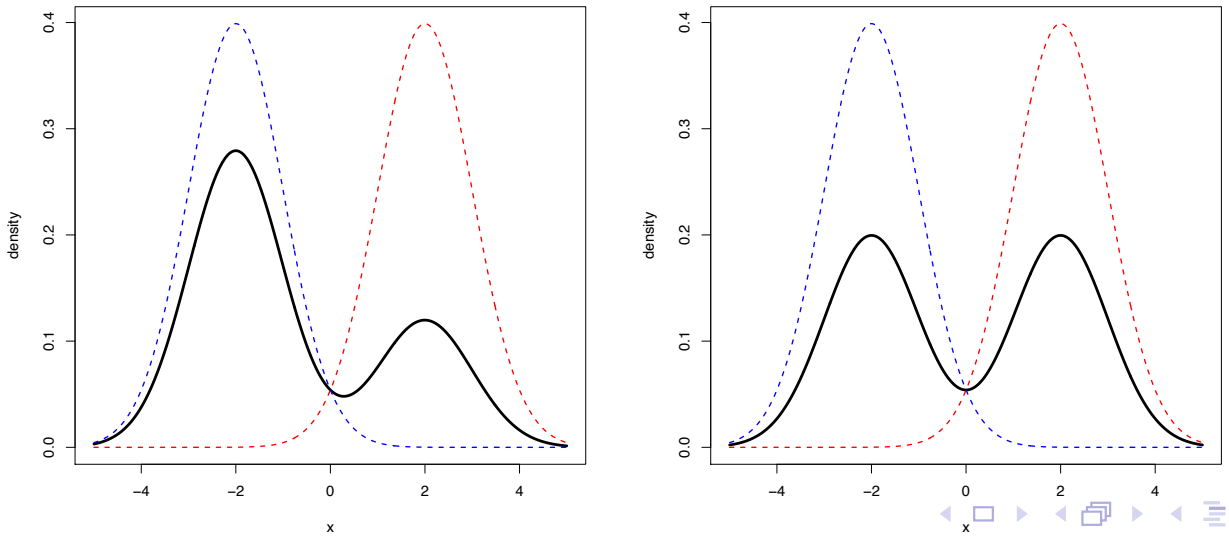
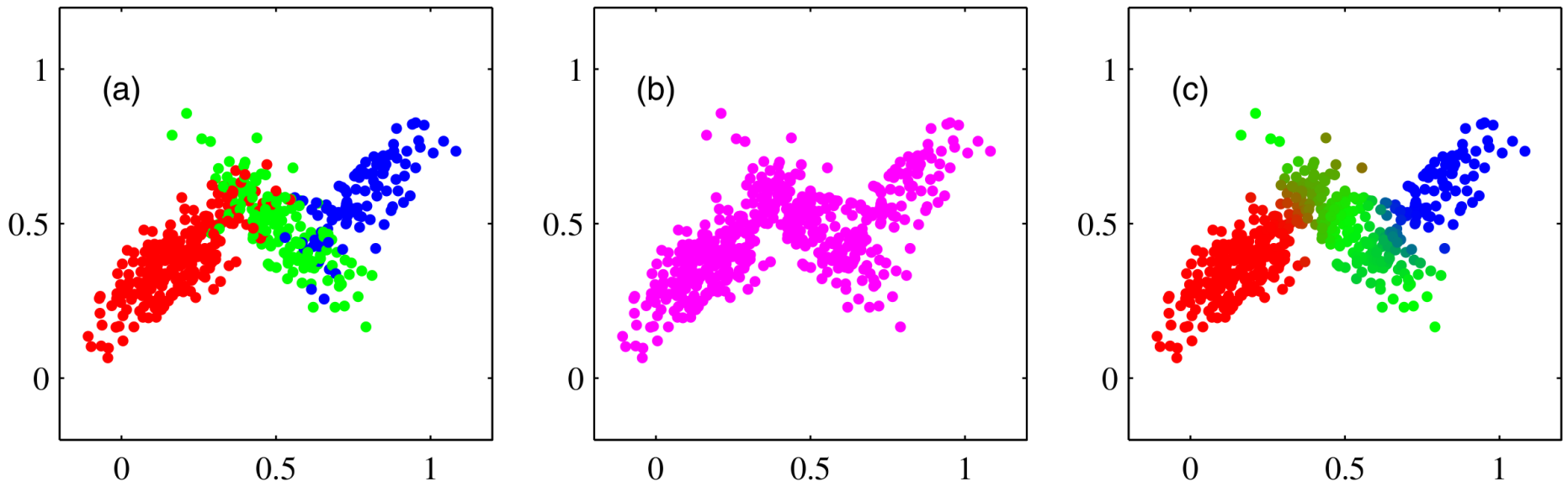


Figure: The density fo two Gaussian components.




(a)
(b)
(c)
1
0.5
0
0
0.5
1

Unknown parameters to infer:

- $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)$, so $\boldsymbol{\pi} \in \mathbb{R}^K$
- $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K)$, for each i, $\boldsymbol{\mu}_i \in \mathbb{R}^n$, so $\boldsymbol{\mu} \in \mathbb{R}^{nK}$
- $\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_K)$, for each i, $\boldsymbol{\Sigma}_i \in \mathbb{R}^{n \times n}$. So $\boldsymbol{\Sigma} \in \mathbb{R}^{n^2 K}$

With a sample of size N: $\{\boldsymbol{x}_i\}_{i=1}^N$, the log-likelihood function takes the form

$$l(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \log \prod_{i=1}^{N} p(\boldsymbol{x}_i | \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{i=1}^{N} \log \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\boldsymbol{x}_i | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}.$$

Direct maximizing l is challanging, since

- the dimension of the problem is high: $(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) \in \mathbb{R}^{K(1+n+n^2)}$;
- l is in general not concave, so there might be many local minimums;
- directly solving $\nabla l = 0$ is generally impossible.

EM for Gaussian Mixture Model Inference

- Initialize the means $\boldsymbol{\mu}_k$'s, the covariance matrices $\boldsymbol{\Sigma}_k$'s, and the mixing coefficients π_k's. Note that $\pi_1 + \ldots + \pi_K = 1$.
- In each iteration, for each $k = 1, \ldots, K$ and $n = 1, \ldots, N$, define

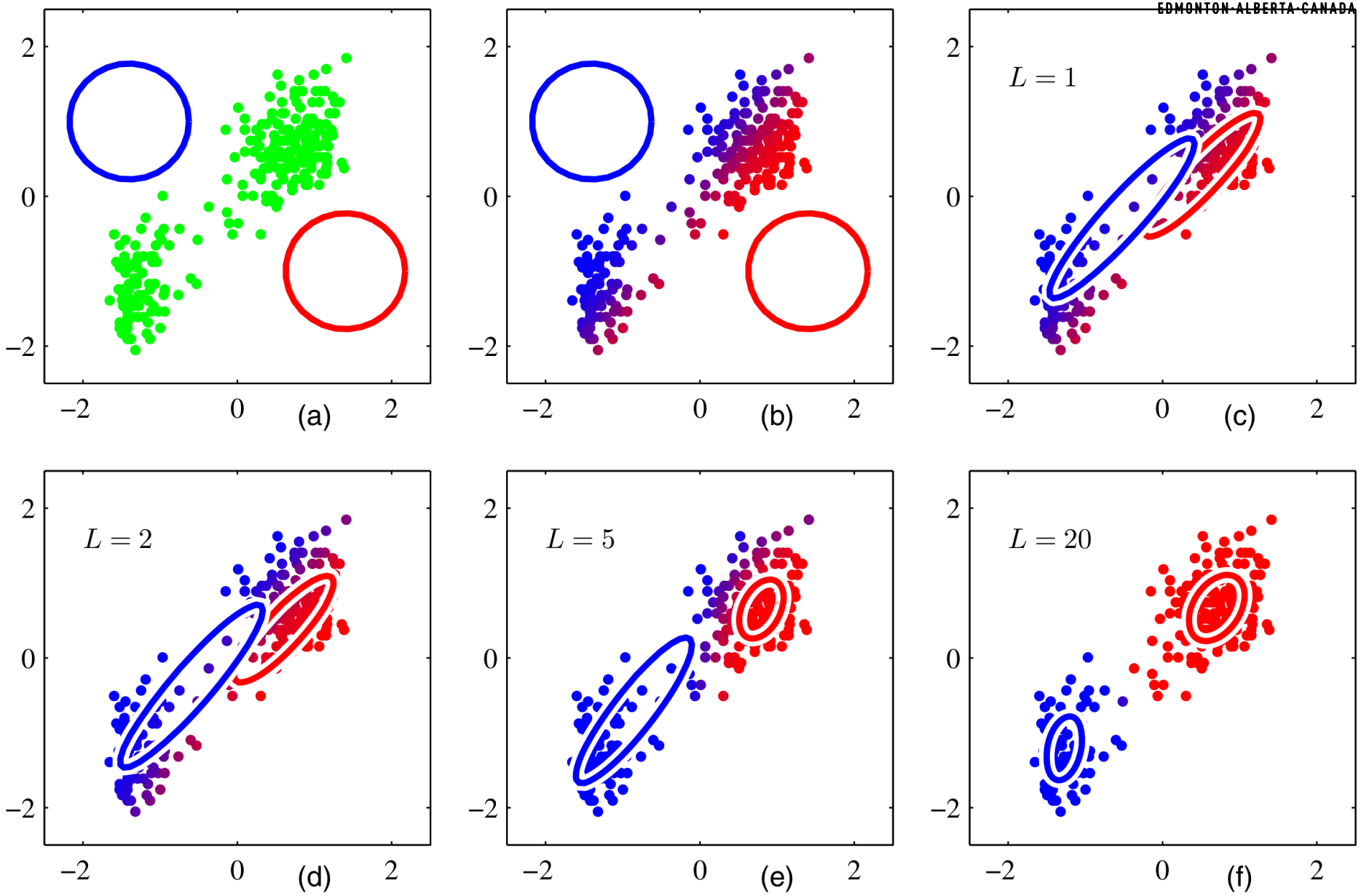
$$\gamma_{n,k} = \frac{\pi_k \mathcal{N}(\boldsymbol{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\boldsymbol{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}, \qquad N_k = \sum_{n=1}^{N} \gamma_{n,k}.$$

- In each iteration: for each $k = 1, \ldots, K$, update:

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma_{n,k} \boldsymbol{x}_n,$$

$$\boldsymbol{\Sigma}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma_{n,k} (\boldsymbol{x}_n - \boldsymbol{\mu}_k^{\text{new}})(\boldsymbol{x}_n - \boldsymbol{\mu}_k^{\text{new}})^T,$$

$$\pi_k^{\text{new}} = \frac{N_k}{N}.$$


(a)
(b)
$L = 1$
(c)
$L = 2$
(d)
$L = 5$
(e)
$L = 20$
(f)

Summary and Remark

- Introduction
- Methods and Algorithm